Arista ELEC-ACC-01 Access Node Routing Protocol Flap Failure Resolution SOP

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_9135

Description

The ELEC-ACC-01 access node, situated at the Electronic City Hub, exhibits a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure domain due to the inherent complexities of the Arista EOS platform's implementation of these protocols. Specifically, the failure is attributed to the dynamic nature of BGP route advertisements, OSPF adjacency establishment, and ISIS LSP flooding, which can lead to transient routing instability. The Arista EOS platform's reliance on software-based forwarding and routing table management further exacerbates the issue, as it can introduce additional latency and processing delays, contributing to the routing protocol flap phenomenon. Moreover, the lack of redundant links and or hardware-based forwarding planes in the ELEC-ACC-01 node exacerbates the issue, rendering the network more susceptible to routing protocol flaps.

Pre-requisite Checks

- Check: Verify that the redundant link to the ELEC-ACC-01 node is functioning correctly and is not contributing to the routing protocol flap issue., Status: Completed

- Check: Ensure that the remote hands have been dispatched to the Electronic City Hub to perform any necessary hardware checks or repairs., Status: Completed

Potential Root Causes

Cause	Probability
Insufficient BGP route dampening configuration	High
OSPf adjacency issue due to hardware or software misconfiguration	Medium
ISIS LSP flooding caused by misconfigured LSP filters	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: show bgp summary BGP table
version is 1, main routing table version 1 Neighbor V AS MsgRcvd
MsgSent TblVer InQ OutQ Up/Down State | Preview -------------------+--
--------+------+---------+--------+--------+---+-------+-----------+--
-------- 10.10.10.10 + 4 65001 1234 1234 1 0 0 00:01:23 Established --
-----------------+----------+------+---------+--------+--------+---+--
-----+-----------+----------, Description: This command will provide
an overview of the BGP neighbor status, including the number of
messages received and sent, the table version, and the state of the
neighbor.
```

```
Command: show ospf neighbor, Expected Output: show ospf neighbor
Neighbor ID Pri State Dead Time Address Interface -----------------
--- ---------- -------- -------------- -------------- 10.10.10.10
0 FULL/DR 00:01:23 10.10.10.10 Ethernet1/1 ----------------- ---
---------- -------- -------------- --------------, Description:
This command will display the OSPF neighbor status, including the
neighbor ID, priority, state, dead time, address, and interface.

Command: show isis database, Expected Output: show isis database ISIS
LSP Database: Type ID LSP seq Age Metric Next-Hop State ------
------------------------------------------------------------------ L2
10.10.10.10 0x80000001 0x80000001 00:01:23 100 10.10.10.10 Up ------
------------------------------------------------------------------ ,
Description: This command will display the ISIS LSP database,
including the LSP type, ID, sequence number, age, metric, next-hop,
and state.
```

Resolution Steps

Step 1: Step: Adjust BGP route dampening configuration, Command: router bgp 65001 timer bgp dampening 10 100 200 dampening 10 100 200 , Description: This step will adjust the BGP route dampening configuration to prevent frequent route flaps. The timer bgp dampening command sets the dampening timer, and the dampening command sets the penalty and reuse values.

Step 2: Step: Verify OSPF adjacency, Command: show ospf neighbor, Description: This step will verify that the OSPF adjacency is established and stable. If the adjacency is not established, troubleshoot the issue and re-run the command.

Step 3: Step: Verify ISIS LSP flooding, Command: show isis database, Description: This step will verify that the ISIS LSP flooding is not occurring. If LSP flooding is detected, troubleshoot the issue and re-run the command.

Post-Check Verification

```
Command: show bgp summary, Expected Output: show bgp summary BGP table
version is 1, main routing table version 1 Neighbor V AS MsgRcvd
MsgSent TblVer InQ OutQ Up/Down State | Preview --------------------+--
--------+------+---------+--------+--------+---+-------+-----------+--
-------- 10.10.10.10 + 4 65001 1234 1234 1 0 0 00:01:23 Established --
-----------------+----------+------+---------+--------+--------+---+--
-----+-----------+----------, Description: This command will verify
that the BGP neighbor status is stable and the routing table is
up-to-date.

Command: show ospf neighbor, Expected Output: show ospf neighbor
Neighbor ID Pri State Dead Time Address Interface -----------------
--- ---------- --------- --------------- --------------- 10.10.10.10
0 FULL/DR 00:01:23 10.10.10.10 Ethernet1/1 ----------------- ---
---------- --------- --------------- ---------------, Description:
This command will verify that the OSPF neighbor status is stable and
the adjacency is established.

Command: show isis database, Expected Output: show isis database ISIS
LSP Database: Type ID LSP seq Age Metric Next-Hop State ------
------------------------------------------------------------------ L2
10.10.10.10 0x80000001 0x80000001 00:01:23 100 10.10.10.10 Up ------
------------------------------------------------------------------ ,
Description: This command will verify that the ISIS LSP database is
stable and there is no LSP flooding.
```

Roll-back Procedures

- Step: Restore original BGP route dampening configuration, Command: router bgp 65001 no timer bgp dampening 10 100 200 no dampening 10 100 200 , Description: This step will restore the original BGP

route dampening configuration to prevent any unintended changes to the routing behavior.

- Step: Verify OSPF adjacency, Command: show ospf neighbor, Description: This step will verify that the OSPF adjacency is established and stable after the rollback.

- Step: Verify ISIS LSP flooding, Command: show isis database, Description: This step will verify that the ISIS LSP flooding is not occurring after the rollback.